

Batches Algorithm for Passive Radar: A Theoretical Analysis

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Passive bistatic radar or passive coherent location is gaining interest in the radar community, as it provides some advantages with respect to active radar. Passive radar does not aim to replace active radar; it provides a good complement to it. The computational effort that is required to implement the required signal processing is one of the drawbacks that affect passive radars. In this paper, a suboptimal but computationally affordable detection algorithm is investigated that is applicable to arbitrary waveforms (different types of illuminators of opportunity). First, a detailed mathematical formulation of the proposed suboptimum algorithm is derived. A theoretical performance analysis is then provided based on a comparison of the proposed with the optimum two-dimensional matched filter. Finally, simulated and real data are used to demonstrate the effectiveness of the proposed algorithm and to validate the theoretical performance analysis.

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I. INTRODUCTION

Passive bistatic radar (PBR) is gradually making progress, both in terms of signal processing and system development (demonstrators and prototypes). Several electromagnetic sources, i.e., illuminators of opportunity, can be used to implement such a radar system. Ground-based broadcast transmitters, such as frequency modulation (FM) and digital audio broadcasting (DAB) radio, as well as analogue and digital television have been largely exploited to demonstrate their capability to perform air and coastal surveillance [1].

In a PBR system, the optimum receiver (from the point of view of both detection and the parameters estimation theory) is implemented by calculating the so-called two-dimensional (2D) cross-ambiguity function (CAF) [2, 3]. In other words, the CAF represents the optimum 2D matched filter that provides the signal processing gain to allow for the detection of targets and estimation of their bistatic range and Doppler shift. The evaluation of the CAF can be computationally expensive as large 2D range-Doppler maps might be required, depending on the extent of the desired surveillance area and the radar resolution, both along the range and the Doppler coordinates. Moreover, a long integration time is usually required to achieve a desirable signal processing gain to allow for the detection of small and/or distant targets. Several suboptimum algorithms have been proposed in the literature to achieve real-time processing capabilities [2, 4–7]. This paper will focus on one of these algorithms, namely, the FM continuous wave (CW) algorithm or batches algorithm, which was presented in [2] and in [7]. Specifically, in [2], a particular example of the FM waveform has been presented, whereas in [7], a more analytical formulation has been proposed. In [8], some preliminary results concerning the generalized formulation of the batches algorithm and the comparison in terms of computational load and signal-to-noise ratio (SNR) losses by using numerical simulations have been presented by the authors. Specifically, in this work, a deep analytical development and a theoretical performance analysis of the algorithm have been defined. Moreover, some parameters have been defined to evaluate the losses in terms of both peak amplitude and side lobe, with respect to the optimum algorithm. Finally, the theoretical performance analysis has been validated by using both real and simulated data. A newly revised formulation of the optimum CAF is proposed in Section II, followed by a new generalized and detailed mathematical formulation of the suboptimum batches algorithm. A theoretical comparative study between the optimum and the suboptimum approach is presented in Section III in terms of SNR and computational load. The theoretical analysis is carried out with the aim of predicting SNR losses when using the proposed algorithm in dependence of target and algorithm parameters. This analysis defines a useful procedure for estimating the performance of the batches algorithm. The proposed theoretical analysis is independent of the type of

waveform. In addition, the newly proposed mathematical formulation of the revised optimum CAF processing and the batches algorithm suggest an interesting comparative analysis between the passive radar processing and the classical active pulsed radar processing. The analogies between the PBR processing and the classical active pulsed radar processing are discussed in Section III. Finally, in Section IV, a case study that considers digital video broadcasting-terrestrial (DVB-T) signals is presented, and both simulated and real data are used to demonstrate the effectiveness of the proposed algorithm and to validate the theoretical performance analysis.

II. REVISED FORMULATION OF THE BATCHES ALGORITHM

A. Problem Formulation

In a passive radar system, the optimum way to process the received signal is given by the theoretical CAF calculation [2, 3], defined as

$$M(\tau, \nu) = \int_0^{T_{\text{int}}} x_{\text{surv}}(t) x_{\text{ref}}^*(t - \tau) e^{-i2\pi \nu t} dt \quad (1)$$

$$0 \leq \tau \leq \tau_{\text{max}} \quad -\nu_{\text{max}} \leq \nu \leq \nu_{\text{max}}$$

where $M(\tau, \nu)$ represents the range-Doppler cross-correlation function between the reference signal $x_{\text{ref}}(t)$ and the surveillance signal $x_{\text{surv}}(t)$, the variable τ denotes the time delay, corresponding to the bistatic time of arrival difference, τ_{max} is the maximum delay of interest, and it is related to the maximum nonambiguous bistatic range, ν corresponds to the frequency Doppler shift, ν_{max} is the maximum Doppler shift of interest, and it is related to the maximum bistatic velocity of interest, T_{int} denotes the integration time or the so-called coherent processing interval. The integration time is typically chosen equal to $T_{\text{int}} = T_{\text{obs}} + \tau_{\text{max}}$, where T_{obs} is the length of the reference signal in order to have no integration losses.

The most obvious way to view the cross-correlation processing would be to calculate the Fourier transform of the signal $x_{\text{surv}}(t) x_{\text{ref}}^*(t - \tau)$, known in literature as the mixing product, for each bistatic time delay. The second way to view the cross-correlation processing would be to calculate the cross correlation between $x_{\text{surv}}(t)$ and $x_{\text{ref}}(t - \tau)$ for each Doppler shift. These two approaches are known in literature as Fourier transform methods and cross correlation, respectively [2]. An efficient implementation of these approaches can be obtained by exploiting the well-known fast Fourier transform (FFT) algorithm. Although the FFT approach can reduce the computational load of the CAF computation, there are several drawbacks to this approach, and these must be underlined [2]. First of all, for long integration time and with the high sampling frequency that may be required when using DAB, DVB-T, worldwide interoperability for microwave access, and universal mobile telecommunication system signal types, the FFT

calculation may be impractical, as the sequences needing to be transformed would be too long. Second, the FFT algorithm takes into account all Doppler frequencies up to half of the sampling frequency. Typically, the maximum Doppler shift is much smaller than the sampling frequency (i.e., $\nu_{\text{max}} \ll f_s$, where f_s is the sampling frequency). Therefore, the majority of the FFT points are discarded after being calculated.

The definition of suboptimum implementations to achieve real-time processing capabilities was discussed in the past by several authors [2–7]. The second issue may be resolved by applying a decimation technique that allows data to be discarded at Doppler frequencies, where it is known that targets do not exist, before calculating the Fourier transform [6]. This paper will focus on one such algorithm, namely, the FMCW algorithm or batches algorithm. A theoretical definition of the batches algorithm is proposed in the next section.

B. Revised Formulation of the Optimum Theoretical CAF

In this section, a revised formulation of the optimum CAF, shown in (1), is proposed.

From (1), the reference signal $x_{\text{ref}}(t)$ can be seen as the sum of n_B contiguous batches of length T_B , and it can be written as

$$x_{\text{ref}}(t) = \sum_{k=0}^{n_B-1} x_i(t - kT_B), \quad (2)$$

where n_B is the number of batches obtained as

$$n_B = \left\lfloor \frac{T_{\text{obs}}}{T_B} \right\rfloor, \quad (3)$$

where $\lfloor \cdot \rfloor$ indicates the maximum integer value that is smaller than the argument. Note that T_{obs} is the length of the reference signal. The signal in each block is defined as

$$x_k(t) = x_{\text{ref}}(t + kT_B) q(t), \quad (4)$$

where $q(t)$ is

$$q(t) = \begin{cases} 1 & t \in [0, T_B] \\ 0 & \text{otherwise} \end{cases}. \quad (5)$$

Substituting (2) into (1), the optimum CAF can be written as

$$M(\tau, \nu) = \sum_{k=0}^{n_B-1} \int_0^{T_{\text{int}}} x_{\text{surv}}(t) x_k^*(t - \tau - kT_B) e^{-i2\pi \nu t} dt. \quad (6)$$

Starting from the following equation,

$$x_k^*(t - \tau - kT_B) \neq 0 \quad \forall t \in [kT_B + \tau, kT_B + \tau + T_B], \quad (7)$$

the integral in (6) can be modified as

$$M(\tau, \nu) = \sum_{k=0}^{n_B-1} \int_{kT_B + \tau}^{kT_B + \tau + T_B} x_{\text{surv}}(t) x_k^*(t - \tau - kT_B) e^{-i2\pi \nu t} dt. \quad (8)$$

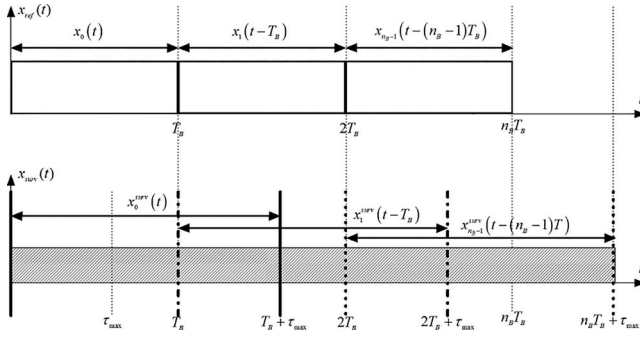


Fig. 1. Reference and surveillance signals segmentation.

Then, with a change of variable $\alpha = t - kT_B$, it now follows that (8) yields:

$$M(\tau, \nu) = \sum_{k=0}^{n_B-1} e^{-i2\pi \nu k T_B} \int_{\tau}^{\tau+T_B} x_{surv}(\alpha + kT_B) \times x_k^*(\alpha - \tau) e^{-i2\pi \nu \alpha} d\alpha. \quad (9)$$

Because τ is supposed to be in the range 0 to τ_{max} , the integration limits, expressed in (9), become 0 and $T_B + \tau_{max}$. In addition, (9) can be reformulated as

$$M(\tau, \nu) = \sum_{k=0}^{n_B-1} e^{-i2\pi \nu k T_B} \int_0^{T_B+\tau_{max}} x_k^{surv}(\alpha) \times x_k^*(\alpha - \tau) e^{-i2\pi \nu \alpha} d\alpha, \quad (10)$$

where $x_k^{surv}(t)$ is the k th block of the surveillance signal, $x_{surv}(t)$, which is defined as

$$x_k^{surv}(t) = x_{surv}(t + kT_B) g(t), \quad (11)$$

where $g(t)$ is

$$g(t) = \begin{cases} 1 & t \in [0, T_B + \tau_{max}] \\ 0 & \text{otherwise} \end{cases}. \quad (12)$$

As it is possible to note from (11) and Fig. 1, the blocks on the surveillance channel are partially overlapped. By considering the signals $x_k^{surv}(t)$ and $x_k(t)$, the CAF $M_k(\tau, \nu)$, calculated within each batch, can be defined as

$$M_k(\tau, \nu) = \int_{-\infty}^{+\infty} x_k^{surv}(t) x_k^*(t - \tau) e^{-i2\pi \nu t} dt. \quad (13)$$

Thus, by substituting (13) into (10) yields

$$M(\tau, \nu) = \sum_{k=0}^{n_B-1} e^{-i2\pi \nu k T_B} M_k(\tau, \nu) \quad (14)$$

$$0 \leq \tau \leq \tau_{max} \quad -\nu_{max} \leq \nu \leq \nu_{max}$$

Therefore, the optimum CAF $M(\tau, \nu)$, between $x_{surv}(t)$ and $x_{ref}(t)$, can be seen as a weighed sum of the CAFs, $M_k(\tau, \nu)$, evaluated within each batch.

Up to this stage, only a new formulation of the optimum CAF has been suggested without gaining, with

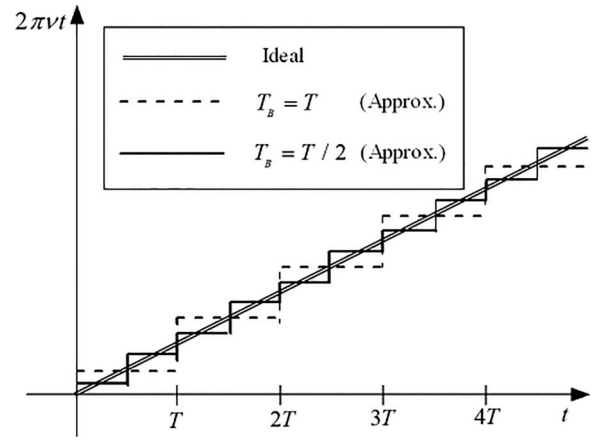


Fig. 2. Phase approximation.

respect to the computational load and time processing. Starting from (14), a description of the proposed suboptimum method follows.

C. Theoretical Definition of the Batches Algorithm

In this section, a mathematical definition of the suboptimum batches algorithm is derived by considering a specific approximation in the Doppler domain. Specifically, in (10), when the product between T_B and the maximum target Doppler ν_{max} is small compared with the unity, the phase rotation can be approximated within each block as a constant value. Specifically, the value of such a phase rotation is closer to the central time of the batch interval ($T_B/2$)

$$2\pi \nu t \approx 2\pi \nu \frac{T_B}{2} \quad \forall t \in [0, T_B], \nu \in [-\nu_{max}, \nu_{max}] \quad (15)$$

and (10) can be simplified as

$$M_b(\tau, \nu) = e^{-i\pi \nu T_B} \sum_{k=0}^{n_B-1} e^{-i2\pi \nu k T_B} \times \int_0^{T_B+\tau_{max}} x_k^{surv}(\alpha) x_k^*(\alpha - \tau) d\alpha, \quad (16)$$

where the subscript “b” stands for “batches algorithm.”

Note that the target Doppler shift is estimated based on the increasing phased shift between consecutive batches (see Fig. 2). Due to the sampling theorem, this yields the relation $\nu_{max} \leq 1/2T_B$. To reduce the losses due to the small Doppler approximation given in (15), $T_B \leq 1/2\nu_{max}\gamma$ is considered, where γ is a constant, which could be defined in relation to an acceptable loss.

Defining the cross correlation within the k th block as

$$x_{cc}^k(\tau) = \int_0^{T_B+\tau_{max}} x_k^{surv}(\alpha) x_k^*(\alpha - \tau) d\alpha \quad (17)$$

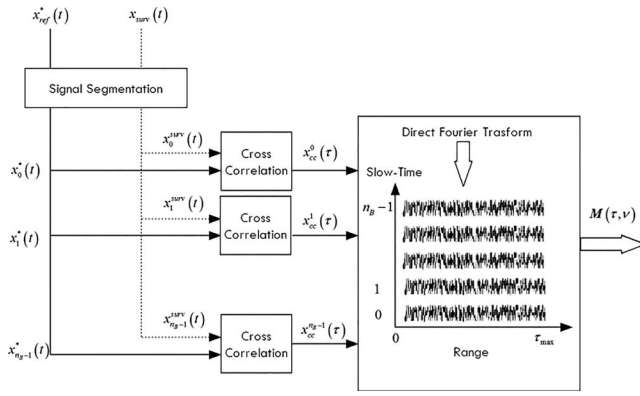


Fig. 3. Reference and surveillance signals segmentation.

(10) can be written as

$$M_b(\tau, \nu) = e^{-i2\pi\nu T_B} \sum_{k=0}^{n_B-1} e^{-i2\pi\nu k T_B} x_{cc}^k(\tau) \quad (18)$$

or alternatively

$$M_b(\tau, \nu) = e^{-i2\pi\nu T_B} \sum_{k=0}^{n_B-1} e^{-i2\pi\nu k T_B} M_k(\tau, 0). \quad (19)$$

The Doppler frequency compensation is neglected inside each block, while it is considered with respect to consecutive blocks.

The main steps of this approach, schematically shown in Fig. 3, can be summarized as the following:

- Select n_B consecutive batches $x_k(t)$ of the reference channel, and n_B partially overlapped batches $x_k^{surv}(t)$ of the surveillance channel.
- Iteratively, for $k = 0, \dots, n_B - 1$ calculate the cross correlation, defined as $x_{cc}^k(\tau)$, between $x_k(t)$ and $x_k^{surv}(t)$ [see (17)].
- The Doppler dimension is then obtained by performing a Fourier transform over the cross-correlation values for each time delay τ or range bin.

III. THEORETICAL PERFORMANCE ANALYSIS—BATCHES ALGORITHM

A. SNR Analysis

Typically, radar performance is evaluated in terms of SNR. Under some typical assumptions, the received echo on the surveillance channel, after downmixing to baseband, can be well approximated by

$$x_{surv}(t) = x_{ref}^s(t) + w(t) = A x_{ref}(t - \tau_T) e^{i2\pi\nu_T t} + w(t), \quad (20)$$

where τ_T , ν_T are the actual time delay and Doppler shift of a single slowly fluctuating point target. The complex constant A represents various effects, such as target reflectivity, propagation effects, and antenna gain. $x_{ref}(t)$ is the signal transmitted by the input/output (IO), $x_{surv}^s(t) = A x_{ref}(t - \tau_T) e^{i2\pi\nu_T t}$ is the target-received echo, and $w(t)$ is an additive random noise process. The

entire CAF calculation problem is linear, so the signal and the noise terms at the output can be separated as

$$M(\tau, \nu) = M^{(s)}(\tau, \nu) + M^{(w)}(\tau, \nu), \quad (21)$$

where $M^{(s)}(\tau, \nu)$ and $M^{(w)}(\tau, \nu)$ represent the useful signal and the noise component at the output of the CAF processing, respectively.

Let us define the optimum SNR as the ratio of the signal power to the expected power in the noise term

$$SNR_{opt}(\tau, \nu) = \frac{|M^{(s)}(\tau, \nu)|^2}{E \left\{ |M^{(w)}(\tau, \nu)|^2 \right\}}, \quad (22)$$

where $M^{(s)}(\tau, \nu)$ and $M^{(w)}(\tau, \nu)$ are, respectively, the target and noise components obtained at the output of the optimum CAF [shown in (14)] and $E \{ \cdot \}$ represents the statistical expectation operator.

On the other hand, the $SNR_b(\tau, \nu)$ at the output of the batches algorithm is calculated as

$$SNR_b(\tau, \nu) = \frac{|M_b^{(s)}(\tau, \nu)|^2}{E \left\{ |M_b^{(w)}(\tau, \nu)|^2 \right\}}, \quad (23)$$

where $M_b^{(s)}(\tau, \nu)$ and $M_b^{(w)}(\tau, \nu)$ represent the target and noise components obtained at the output of the batches algorithm, respectively, as shown in (18).

The target components $M^{(s)}(\tau, \nu)$ and $M_b^{(s)}(\tau, \nu)$, obtained, respectively, at the output of the optimum and batches algorithm, are calculated by substituting the target component $x_{surv}^s(t)$ into (14) and (18). This yields (see Appendix)

$$\begin{cases} M^{(s)}(\tau, \nu) = A e^{-i2\pi(\nu - \nu_T)\tau_T} \\ \quad \times \sum_{k=0}^{n_B-1} e^{-i2\pi(\nu - \nu_T)k T_B} \chi_k(\tau - \tau_T, \nu - \nu_T) \\ M_b^{(s)}(\tau, \nu) = A e^{i2\pi\nu_T \tau_T} \\ \quad \times \sum_{k=0}^{n_B-1} e^{-i2\pi(\nu - \nu_T)k T_B} \chi_k(\tau - \tau_T, -\nu_T) \end{cases}, \quad (24)$$

where $\chi_k(\tau, \nu)$ is the ambiguity function of the reference signal belonging to the k th batch, defined as

$$\chi_k(\tau, \nu) = \int_{-\infty}^{+\infty} x_k(t) x_k^*(t - \tau) e^{-i2\pi\nu t} dt. \quad (25)$$

The noise components $M^{(w)}(\tau, \nu)$ and $M_b^{(w)}(\tau, \nu)$ obtained, respectively, at the output of the optimum and batches algorithm, are calculated by substituting the noise component $w(t)$ into (14) and (18). This yields (see

$$\begin{cases} E \left\{ |M^{(w)}(\tau, \nu)|^2 \right\} = \sum_{k=0}^{n_B-1} \int S_w(f) |X_k(f - \nu)|^2 df \\ E \left\{ |M_b^{(w)}(\tau, \nu)|^2 \right\} = \sum_{k=0}^{n_B-1} \int S_w(f) |X_k(f)|^2 df \end{cases}, \quad (26)$$

where $X_k(f)$ is the Fourier transform of the reference signal $x_k(t)$ in the k th batch and $S_w(f)$ is the noise power spectral density.

From (26) and under the assumption of additive white noise (AWN), the output noise power can be considered the same for both the batches and optimum algorithm. Assuming an additive coloured noise, the equality $\int S_w(f) |X_k(f - \nu)|^2 df = \int S_w(f) |X_k(f)|^2 df$ is a good approximation when the maximum considered Doppler $\nu_{\max}$ is small, with respect to the bandwidth of the single-batch transmitted signal. Under this assumption, the output noise power can be considered the same for both the batches and the optimum algorithms.

Furthermore, to evaluate the losses of the batches algorithm, with respect to the optimum algorithm, the improvement factor (IF) can be defined

$$IF(\tau, \nu) = 20 \log_{10} \left(\left| \frac{SNR_b(\tau, \nu)}{SNR_{opt}(\tau, \nu)} \right| \right), \quad (27)$$

where $SNR_b(\tau, \nu)$ and $SNR_{opt}(\tau, \nu)$ represent the SNR obtained, respectively, by using the batches algorithm and the optimum one.

From (27), the results obtained during the previous analysis equation yields

$$\begin{aligned} IF(\tau, \nu) &= 20 * \log_{10} \left(\left| \frac{M_b^{(s)}(\tau, \nu)}{M^{(s)}(\tau, \nu)} \right| \right) \\ &= 20 * \log_{10} \left(\left| \frac{\sum_{k=0}^{n_B-1} e^{-i2\pi(\nu-\nu_T)kT_B} \chi_k(\tau - \tau_T, -\nu_T)}{\sum_{k=0}^{n_B-1} e^{-i2\pi(\nu-\nu_T)kT_B} \chi_k(\tau - \tau_T, \nu - \nu_T)} \right| \right). \end{aligned} \quad (28)$$

From (28), the ratio between the target peak evaluated with both the batches algorithm and the optimum one is

$$\begin{aligned} PeakRatio(\tau_T, \nu_T) &= 20 * \log_{10} \left(\left| \frac{M_b^{(s)}(\tau_T, \nu_T)}{M^{(s)}(\tau_T, \nu_T)} \right| \right) \\ &= 20 * \log_{10} \left(\left| \frac{\sum_{k=0}^{n_B-1} \chi_k(0, -\nu_T)}{\sum_{k=0}^{n_B-1} \chi_k(0, 0)} \right| \right). \end{aligned} \quad (29)$$

TABLE I
Computational Cost

Algorithm	Complex Multiplication
Cross-correlation FFT approach	$2N \log_2(N) + [1 + \log_2(N)] N N_{Doppler}$
Fourier transform FFT approach	$[1 + \log_2(N)] N N_{bin}$
Batch algorithm	$[3 \log_2(N_{bin}) + 1 + \log_2(n_B)] N_{bin} n_B$

In conclusion, the losses only depend on the target Doppler frequency ν_T and on the shape of the ambiguity function $\chi_k(0, \nu)$, evaluated within each batch, along the Doppler coordinate.

Specifically, if the small Doppler approximation in (15) can be accepted, the losses can be considered negligible. Moreover, the small Doppler approximation is acceptable if the product $\nu_{\max} T_B$ is small and for this reason, the batch length T_B can be shortened as required to obtain reduced losses. The performance will depend on the type of waveform, as some waveforms are intrinsically less influenced by the Doppler effect than others. Therefore, the optimal selection of the batch length depends, in general, on the type of waveform. Note that (28) can be used to predict the algorithm performance, with respect to a given IO and target velocity. Specifically, (28) can be evaluated by using a closed-form expression of the ambiguity function $\chi_k(\tau, \nu)$, if it is available, or by computing $\chi_k(\tau, \nu)$, starting from the reference signal.

B. Computational Cost Analysis

An efficient implementation of lossless approaches, such as correlation FFT and direct FFT, can be obtained by exploiting the well-known FFT algorithm [2], as mentioned in Section II.A. As discussed in [2], when exploiting an FM illuminator of opportunity, and in [7, 8], when DVB-T signals are used, the processing time can be strongly reduced by using the batches algorithm. Specifically, in [8], it was shown that it is possible to reduce the processing time by more than 96%, with respect to the optimum case. The computational load in terms of complex multiplication relative to the batches algorithm is shown in Table I, with respect to the two lossless approaches. In particular, N is the number of samples, N_{bin} and $N_{Doppler}$ represent the number of range and Doppler bins included in the CAF, respectively. The computational load for the batches algorithm increases when the number of batches n_B and the number N_{bin} of range bins included in the CAF increase. In other words, if the batch length T_B decreases, the number of batches n_B increases, and consequently, the computational effort becomes more expensive due to the higher number of cross-correlation computations.

In conclusion, the batch length T_B can be set to obtain a good compromise between SNR losses and computational cost. While the processing is not optimum, it is possible to obtain a real-time CAF calculation with a

general-purpose personal computer (PC). Furthermore, the losses should not significantly affect the detection performances with a suitable batch length selection.

C. Analogy with Pulse-Doppler Radar

It is important to point out how the optimal theoretical CAF and the batches algorithm present analogies with the classical processing used in active pulse-Doppler radars. The common method typically used to process the received signal in an active pulse-Doppler radar is pulse-Doppler processing [9]. The received signal is first pulse compressed by correlating the fast-time signals with the zero-Doppler matched filter. The range-Doppler map is then formed by taking the Fourier transform along the slow time domain. This processing is optimum under the assumption that the received waveform is an amplitude-scaled and time-delayed version of the transmit waveform. The Doppler shift of the received waveform represents an unintentional mismatch between the received signal and the matched filter. The response of a waveform in the presence of an uncompensated Doppler is known in literature as “Doppler tolerance.” In an active radar system, the Doppler shift and its impact on the matched filter output signal may be approached in several ways. The first way is to select a waveform and its modulation parameters to achieve a Doppler-tolerant waveform. Second, if it is possible to estimate the target’s radial velocity, the received signal can be countered by applying an equal Doppler frequency shift to the received signal. Finally, a bank of matched filters, each one tuned to a different Doppler shift, may be used. This method can be considered as the optimum receiver in an active pulse-Doppler radar. Pulse-Doppler processing is perhaps the most commonly used method because it can often provide a significant reduction in computational cost, while suffering only a modest loss in processing gain.

Comparing these two mentioned approaches with the optimal theoretical CAF and the batches algorithm, it is possible to conclude that there are the following analogies between the CAF calculation in a passive radar system and the typical processing adopted in active radar systems: the pulse repetition interval T_R is the equivalent of the batch length T_B ; the transmitted signal for each pulse $z(t)$ is the equivalent of the reference signal belonging to each batch $x_k(t)$; and the received signal $r_k(t)$ is the equivalent of $x_k^{surv}(t)$. Specifically, the matched filter bank approach in an active pulse-Doppler radar, which implements a matched filter in both time delay and Doppler, is based on the same theoretical concept as the optimum CAF approach in a passive scenario. Both the pulse-Doppler and batches algorithm have been developed to reduce the computational load, and their performances are strictly related to the Doppler tolerance of the transmitted waveform. Similar to the passive radar scenario, the radar performances are evaluated in terms of SNR, and the performances between the pulse-Doppler processing and the bank of matched filters are compared. Therefore, as

shown in the case of passive radars, the following parameter is defined

$$IF_{PD}(\tau, \nu) = 20 \log_{10} \left(\left| \frac{SNR_{PD}(\tau, \nu)}{SNR_{MF}(\tau, \nu)} \right| \right), \quad (30)$$

where $SNR_{PD}(\tau_T, \nu_T)$ and $SNR_{MF}(\tau_T, \nu_T)$ represent the SNR obtained by using the pulse-Doppler processing and the bank of matched filters, respectively.

Similar to the passive radar analysis, under the assumption of AWN, it can be shown that

$$IF_{PD}(\tau, \nu) = 20 * \log_{10} \left(\frac{|\chi_Z(\tau - \tau_T, -\nu_T)| \left| \sum_{k=0}^{M-1} e^{-i2\pi(\nu - \nu_T)kT_R} \right|}{|\chi_Z(\tau - \tau_T, \nu - \nu_T)| \left| \sum_{k=0}^{M-1} e^{-i2\pi(\nu - \nu_T)kT_R} \right|} \right), \quad (31)$$

where $\chi_Z(\tau, \nu)$ represents the ambiguity function of the transmitted pulse $z(t)$.

Comparing (31) and (28), it is possible to conclude that the two expressions are very similar. In (28), by assuming $\chi_k(\tau, \nu) = \chi_Z(\tau, \nu) \forall i$, the same result is obtained. The main difference is that in the active radar case, the transmitted waveform properties can be selected to obtain a Doppler-tolerant waveform, while in a passive radar scenario, only an appropriate batch length can be selected to reduce the losses.

IV. CASE STUDY: DVB-T WAVEFORMS

In this section, a specific case using DVB-T waveforms is considered. The theoretical performance analysis developed in Section III is evaluated via simulation, and SNR losses are predicted in an operative scenario when using real data.

A. Simulated Results

Section III demonstrated how the losses of the target’s echo power are negligible along the range coordinate. Therefore, in this section, the losses, with respect to both the batch length and the target’s Doppler for a fixed target range, are explored.

1) *Simulation Setup:* The DVB-T transmitted signal is simulated as

$$x_{ref}(t) = \sum_k \sum_{n=0}^{N-1} s_{OFDM}(k, t) q(t - kT_S), \quad (32)$$

where $q(t)$ is defined as

$$q(t) = \begin{cases} 1 & t \in [-T_G, T_U] \\ 0 & otherwise \end{cases}. \quad (33)$$

In the previous equations, T_U represents the symbol duration interval, T_G represents the guard interval, and $T_S = T_G + T_U$. In this multicarrier modulation scheme, the N transmitted frequencies are orthogonal over a temporal window of length T_U . Considering the complex

TABLE II
DVB-T Signal Specification

Parameter	Values
Subcarrier spacing	1.1 KHz
Subcarriers numbers	6817
Modulation	64 QAM
Bandwidth	7.61 MHz
Blocks per frame	68
OFDM symbol duration	1.12 ms
Useful symbol duration	$T_u = 896 \mu s$
Guard interval duration	$T_u/4$

OFDM, Orthogonal frequency division multiplexing.

TABLE III
Simulation Parameters

Parameter	Values
Sampling frequency	$f_s = 9.14 \text{ MHz}$
Observation time	$T_{obs} = 250 \text{ ms}$
Target Doppler frequency	$\nu_T = [0, 100, 200, 300, 400, 500, 600, 700, 800, 900] \text{ Hz}$
Batch time	$T_B = [15.64, 31.28, 109.38, 218.76, 333.29, 462, 924] \mu s$

elements $s_{k,n}$ belonging to a finite alphabet and representing the transmitted digital data signal, the signal becomes

$$s_{OFDM}(k, t) = \sum_{n=0}^{N-1} s_{k,n} e^{i2\pi \frac{n}{T_u} t}. \quad (34)$$

Next, results have been obtained by considering a 8K-mode DVB-T transmitted signal. The DVB-T signal standard is specified in [10]; for convenience, most parameters are reproduced in Table II.

The received echo on the surveillance channel, after downmixing to baseband, is simulated according to (20). The other parameters used during the simulations are summarized in Table III.

2) *Results:* As previously mentioned, the losses depend on the target Doppler frequency ν_T and on the shape of the ambiguity function $\chi_k(0, \nu)$, evaluated within each batch, with respect to the Doppler domain.

For instance, in the case of a DVB-T signal, the zero range slice of the AF $\chi_k(0, \nu)$ can be viewed as a “sinc” function $\text{sinc}(T_B \nu)$, where T_B is the batch length [7, 11]. In Fig. 4, the zero range slice $|\chi_k(0, \nu)|$, averaged on different batches, is calculated. The different lines refer to the considered batch lengths.

It is possible to conclude that if the batch length is small, the zero path delay of the DVB-T ambiguity function is very flat, and consequently, the losses are very low, even for large values of the target Doppler.

On the contrary, when the target’s Doppler frequency and the batch length increase, the target’s echo power decreases, as shown in Fig. 5. Specifically, Fig. 5 shows the peak ratio defined in (29), for the considered target

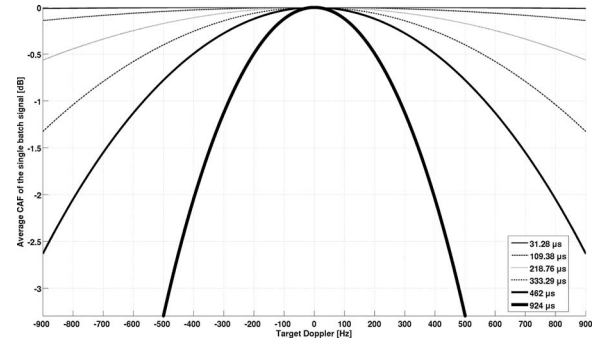


Fig. 4. Zero path delay AF of DVB-T signal.

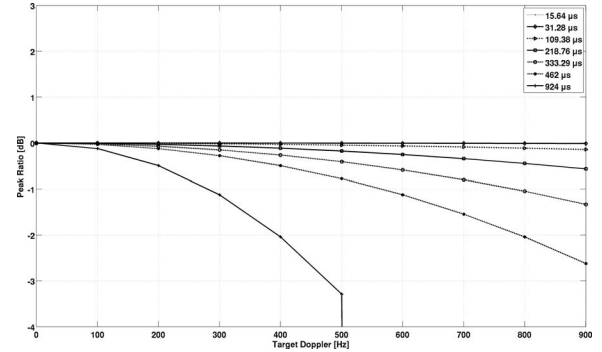


Fig. 5. Peak ratio with respect to target’s Doppler.

Doppler. Due to the sampling theorem, the maximum target’s Doppler is $\nu_{\max} = 1/2T_B$, as mentioned in Section II.B. For a batch length of $924 \mu s$, the maximum target’s Doppler is about 541 Hz. The line relative to this batch length is interrupted in Fig. 5 at the target’s Doppler frequencies $> 500 \text{ Hz}$.

From (28), the following metrics can be defined to evaluate the losses, with respect to the CAF sidelobe levels:

- Integrated sidelobe-level target doppler profile: the ratio between the target’s echo Doppler slice of the CAF evaluated with the batches algorithm and with the optimum approach

$$IF(\tau, \nu_T) = 20 * \log_{10} \left(\frac{\sum_{k=0}^{n_B-1} \chi_k(\tau - \tau_T, -\nu_T)}{\sum_{k=0}^{n_B-1} \chi_k(\tau - \tau_T, 0)} \right). \quad (35)$$

An estimate of the CAF background power is obtained by identifying a region $B(\tau)$ outside the main lobe of the target and by averaging the power in this region. The ratio between the normalized background power has been evaluated by using both the batches and optimum

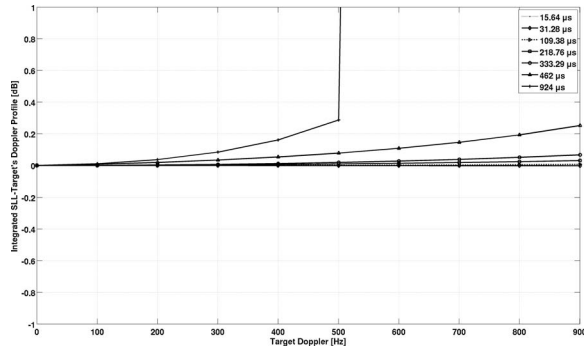


Fig. 6. Sidelobe-level target Doppler profile, with respect to target's Doppler.

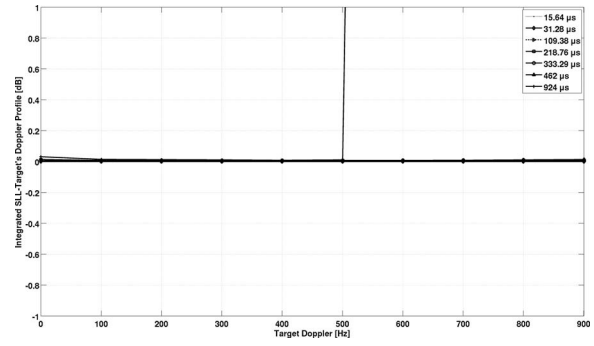


Fig. 7. Integrated sidelobe-level target range profile, with respect to target's Doppler.

approach and have been defined as

$$BP(\tau, v_T) = 10 * \log_{10} \left(\frac{\int_{B(\tau)} |M_b^{(s)}(\tau, v_T)|^2 d\tau}{\int_{B(\tau)} |M^{(s)}(\tau, v_T)|^2 d\tau} \left| \frac{M^{(s)}(\tau_T, v_T)}{M_b^{(s)}(\tau_T, v_T)} \right|^2 \right). \quad (36)$$

- Integrated sidelobe-level target range profile: the ratio between the CAF range slices obtained with the batches and the optimum algorithm

$$IF(\tau_T, v) = 20 * \log_{10} \left(\frac{\sum_{k=0}^{n_B-1} e^{-i2\pi(v-v_T)kT_B} \chi_k(0, -v_T)}{\sum_{k=0}^{n_B-1} e^{-i2\pi(v-v_T)kT_B} \chi_k(0, v - v_T)} \right). \quad (37)$$

Also, in this case, an estimate of the CAF background power is made by identifying a region $B(v)$ outside the main lobe of the target and by averaging the power in this region

$$BP(\tau_T, v) = 10 * \log_{10} \left(\frac{\int_{B(v)} |M_b^{(s)}(\tau_T, v)|^2 dv}{\int_{B(v)} |M^{(s)}(\tau_T, v)|^2 dv} \left| \frac{M^{(s)}(\tau_T, v_T)}{M_b^{(s)}(\tau_T, v_T)} \right|^2 \right). \quad (38)$$

The results in terms of the two defined metrics [see (36) and (38)] are, respectively, shown in Figs. 6 and 7. It is possible to conclude that the losses, with respect to the CAF background, are negligible.

The computational load in terms of complex multiplication is shown in Table IV. Specifically, the

TABLE IV
Computational Cost

Algorithm	Complex Multiplication
Cross-correlation FFT approach	$127 \cdot 10^8$
Fourier transform FFT approach	$505 \cdot 10^8$
Batch algorithm ($T_B = 15.64 \mu s$)	$7.16 \cdot 10^8$
Batch algorithm ($T_B = 31.28 \mu s$)	$3.50 \cdot 10^8$
Batch algorithm ($T_B = 109.38 \mu s$)	$0.96 \cdot 10^8$
Batch algorithm ($T_B = 218.76 \mu s$)	$0.46 \cdot 10^8$
Batch algorithm ($T_B = 333.29 \mu s$)	$0.30 \cdot 10^8$
Batch algorithm ($T_B = 462 \mu s$)	$0.21 \cdot 10^8$
Batch algorithm ($T_B = 924 \mu s$)	$0.10 \cdot 10^8$

computational load has been evaluated by considering the theoretical analysis presented in Table I and the following observations:

- 1) The number of Doppler bins $N_{Doppler}$ is equal to 250, considering an integration time of 250 ms.
- 2) A maximum Doppler of 500 Hz.
- 3) The number of range bins is 1000.

The computational load with the largest batch length is reduced by a factor 18 or 72, with respect to the cross-correlation FFT approach and the Fourier transform FFT approach, respectively.

B. Real Data Results

- 1) *Real Data Setup:* The experimental data were acquired in three different measurement campaigns:

- 1) The first campaign was supported by European Defence Agency (EDA) in the context of the Array Passive ISAR Adaptive Processing Project [12, 13]. In particular, the acquisition system (RX) was located in Paracuellos near Madrid-Barajas airport, whereas the DVB-T transmitter was located in Torrespana. Three DVB-T adjacent channels centered at the frequency of 850 MHz, with a total bandwidth of about 25 MHz, are considered. The integration time was 0.25 s. The bistatic baseline was about 15 km, and the target was a landing plane.
- 2) The receiver was located at the Department of Information Engineering of the University of Pisa, and

TABLE V
Targets Parameters

Data Set	Target Type	Range	Doppler Frequency
1	Plane	2500 m	-176 Hz
1	Plane	3000 m	268 Hz
1	Plane	4000 m	460 Hz
2	Plane	1700 m	-384 Hz
2	Plane	950 m	-344 Hz
3	Ship	3700 m	0 Hz
3	Ship	7400 m	40 Hz
3	Ship	2500 m	-24 Hz
3	Ship	13 000 m	-36 Hz

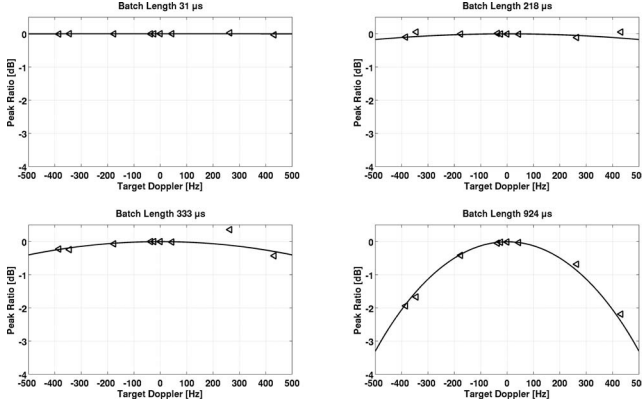


Fig. 8. Comparison between simulation and real data.

the illuminator of opportunity was a DVB-T transmitter located on “Monte Serra” in Pisa [14]. Specifically, the DVB-T transmitter was 14 km away from the receiver. The surveillance antenna was pointed at 17° , with respect to north (i.e., azimuth) and tilted upwards towards the sky at an angle of 30° . The target was a plane.

- 3) The receiver was located at the “Centro di Sperimentazione e Supporto Navale - Istituto per le Telecomunicazioni e l'Elettronica (CSSN-ITE) G. Vallauri” Institute in Livorno and the illuminator of opportunity was a DVB-T transmitter located on “Monte Serra” in Pisa (around 32 km from the receiver), and the surveillance antenna was directed towards an area of the sea in front of the receiver site [15]. The targets were ships.

2) *Results:* Nine targets with different Doppler frequency values from the described data sets are selected. Specifically, the target types and their Doppler frequencies are summarized in Table V.

The comparison between simulations and real data is shown in Fig. 8. Specifically, the ratio between the target's signal peak with both the batches and the optimum algorithm is evaluated according to (29). First, the peak ratio for the simulated signal (solid lines) has been evaluated by considering four batch lengths. Then, the peak ratio, with respect to the real data (left-pointing

TABLE VI
Summarized Results

Batch Length	Peak Ratio $v_T = -176$ Hz	Peak Ratio $v_T = 460$ Hz	Processing Time
$(T_B = 31.28 \mu s)$	-0.0005	-0.0303	4.93 s
$(T_B = 218.76 \mu s)$	-0.0061	0.0536	0.92 s
$(T_B = 333.29 \mu s)$	-0.0699	-0.4297	0.71 s
$(T_B = 924 \mu s)$	-0.4171	-2.19	0.59 s

triangle), has been calculated. It is important to point out that the values obtained with the real data closely match those obtained via simulation. The mismatch between simulation and real data amounts to a maximum absolute value of 0.5 dB, because the theoretical development has been obtained under the assumption of a slowly fluctuating target in AWN environment. In a real scenario, due to the presence of interference, such as direct signal and clutter, the assumption of AWN is no longer valid.

The results presented in Fig. 8 are summarized in Table VI. The processing time has been evaluated with a general-purpose 4-core PC equipped with 16 GB of random-access memory.

Specifically, as shown in Fig. 8, when the target Doppler frequency and the batch length increase, the target power decreases. The processing time is related to the batch length, if the batch length T_B decreases, the computational effort becomes more expensive.

These results demonstrate how the theoretical analysis can be used as a performance prediction tool. Moreover, this analysis allows a batch length to be set to provide a good compromise between losses and processing time.

V. CONCLUSION

A novel formulation of the CAF processing for passive radar systems, namely, the batches algorithm has been proposed. A detailed mathematical analysis of the algorithm has been developed, and a theoretical comparative study between the optimum and suboptimum algorithms has been presented in terms of SNR losses. An interesting comparative analysis between the proposed passive radar processing and the classical active pulsed radar processing has also been presented. Finally, the study has been supported by a comparison between the simulated and real data results, relative to a DVB-T IO. The analysis of the results has confirmed the accuracy of the SNR losses predicted by the theory formulated.

APPENDIX: TARGET COMPONENTS DERIVATION

The target components $M^{(s)}(\tau, \nu)$ and $M_b^{(s)}(\tau, \nu)$ obtained, respectively, at the output of the optimum and batches algorithm, are calculated starting from (14) and (18).

First, by substituting the target component $x_{surv}^s(t)$, shown in (20), into (13), it can be seen that the target component of the CAF $M_k^{(s)}(\tau, \nu)$, evaluated within each batch, is given by

$$M_k^{(s)}(\tau, \nu) = \begin{cases} \sum_{m=1}^M \int_0^{\tau_T} x_{k-m}(t - \tau_T) x_k^*(t - \tau) e^{-i2\pi(\nu - \nu_T)t} dt + \chi_k(\tau - \tau_T, \nu - \nu_T) & 0 \leq \tau \leq \tau_T \\ \chi_k(\tau - \tau_T, \nu - \nu_T) + \sum_{l=1}^L \int_{T_B + \tau_T}^{T_B + \tau_{\max}} x_{k+l}(t - \tau_T) x_k^*(t - \tau) e^{-i2\pi(\nu - \nu_T)t} dt & \tau_T \leq \tau \leq \tau_{\max} \end{cases}, \quad (39)$$

where $\chi_k(\tau, \nu)$ is the ambiguity function of the reference signal belonging to the k th batch, defined as

$$\chi_k(\tau, \nu) = \int_{-\infty}^{+\infty} x_k(t) x_k^*(t - \tau) e^{-i2\pi \nu t} dt. \quad (40)$$

Second, by substituting the target component $x_{surv}^s(t)$, shown in (20), into (17), it can be seen that the target component of the cross correlation $x_{cc}^{k(s)}(\tau)$, evaluated within each batch, is given by

$$x_{cc}^{k(s)}(\tau) = M_k^{(s)}(\tau, 0) = \begin{cases} \sum_{m=1}^M \int_0^{\tau_T} x_{k-m}(t - \tau_T) x_k^*(t - \tau) e^{i2\pi \nu_T t} dt + \chi_k(\tau - \tau_T, -\nu_T) & 0 \leq \tau \leq \tau_T \\ \chi_k(\tau - \tau_T, -\nu_T) + \sum_{l=1}^L \int_{T_B + \tau_T}^{T_B + \tau_{\max}} x_{k+l}(t - \tau_T) x_k^*(t - \tau) e^{i2\pi \nu_T t} dt & \tau_T \leq \tau \leq \tau_{\max} \end{cases}, \quad (41)$$

where $\chi_k(\tau, \nu)$ is the ambiguity function of the reference signal belonging to the k th batch, defined in (40). Note that in (39) and (41), the cross terms between two adjacent batches (see the sums with respect to m and l) are equal to zero in the case $[0 \leq \tau \leq \tau_T; k = 0]$ and $[\tau_T \leq \tau \leq \tau_{\max}; k = n_B]$. Note that the signals belonging to the two different batches can be considered uncorrelated; thus, the cross terms can be ignored.

Under the previous assumption and by substituting (39) into (14), the target component $M^{(s)}(\tau, \nu)$ obtained at the output of the optimum CAF can be expressed as

$$M^{(s)}(\tau, \nu) = e^{-i2\pi(\nu - \nu_T)\tau_T} \sum_{k=0}^{n_B-1} e^{-i2\pi(\nu - \nu_T)kT_B} \times \chi_k(\tau - \tau_T, \nu - \nu_T), \quad (42)$$

where $\chi_k(\tau, \nu)$ is the ambiguity function of the reference signal belonging to the k th batch, as defined in (40).

On the other hand, starting from the same assumption and by substituting (41) into (18), it can be shown that the output target component $M_b^{(s)}(\tau, \nu)$ obtained with the batch algorithm is

$$M_b^{(s)}(\tau, \nu) = e^{i2\pi \nu_T \tau_T} \sum_{k=0}^{n_B-1} e^{-i2\pi(\nu - \nu_T)kT_B} \chi_k(\tau - \tau_T, -\nu_T), \quad (43)$$

where $\chi_k(\tau, \nu)$ is the ambiguity function of the reference signal belonging to the k th batch, as defined in (40).

After that, the noise components $M^{(w)}(\tau, \nu)$ and $M_b^{(w)}(\tau, \nu)$ obtained, respectively, at the output of the

optimum and batches algorithm, are calculated, starting from (14) and (18).

First, by substituting the noise component $w(t)$, shown in (20), into (13), it can be seen that the noise component $M^{(w)}(\tau, \nu)$ is given by

$$M^{(w)}(\tau, \nu) = \sum_{k=0}^{n_B-1} e^{-i2\pi \nu k T_B} M_k^{(w)}(\tau, \nu), \quad (44)$$

where $M_k^{(w)}(\tau, \nu)$ is defined as

$$M_k^{(w)}(\tau, \nu) = \int_{-\infty}^{+\infty} w_k^{surv}(\alpha) x_k^*(\alpha - \tau) e^{-i2\pi \nu \alpha} d\alpha \quad (45)$$

and $w_k^{surv}(t)$ is

$$w_k^{surv}(t) = w(t + kT_B) g(t), \quad (46)$$

where $g(t)$ is

$$g(t) = \begin{cases} 1 & t \in [0, T_B + \tau_{\max}] \\ 0 & \text{otherwise} \end{cases}. \quad (47)$$

The output noise power is

$$\begin{aligned} E \left\{ |M^{(w)}(\tau, \nu)|^2 \right\} &= \sum_{k=0}^{n_B-1} \sum_{l=0}^{n_B-1} e^{-i2\pi(k-l)\nu T_B} \\ &\quad \times E \left\{ M_k^{(w)}(\tau, \nu) M_l^{(w)*}(\tau, \nu) \right\} \\ &= \sum_{k=0}^{n_B-1} E \left\{ |M_k^{(w)}(\tau, \nu)|^2 \right\}, \end{aligned} \quad (48)$$

where

$$E \left\{ \left| M_k^{(w)}(\tau, \nu) \right|^2 \right\} = \int S_w(f) |X_k(f - \nu)|^2 df \quad (49)$$

and $E \{ \cdot \}$ represents the expectation operator and $X_k(f)$ is the Fourier transform of the reference signal $x_k(t)$ belonging to k th batch. Note that (48) was obtained by assuming $E \{ x_k^{(w)}(\tau) x_l^{(w)*}(\tau) \} = 0 \quad \forall l \neq k$.

Whereas by replacing the noise component $w(t)$, shown in (20), with (18), the output noise component $M_b^{(w)}(\tau, \nu)$ obtained with the batch algorithm is

$$M_b^{(w)}(\tau, \nu) = \sum_{k=0}^{n_B-1} e^{-i2\pi\nu k T_B} x_{cc}^{k(w)}(\tau), \quad (50)$$

where the noise range profile $x_{cc}^{k(w)}(\tau)$ of the k th batch is defined as

$$x_{cc}^{k(w)}(\tau) = \int_{-\infty}^{+\infty} w_k^{surv}(\alpha) x_k^*(\alpha - \tau) d\alpha \quad (51)$$

and $w_k^{surv}(t)$ is defined in (46).

The power of the output sample evaluated over the range τ and for the k th batch is

$$E \left\{ \left| x_{cc}^{k(w)}(\tau) \right|^2 \right\} = \int S_w(f) |X_k(f)|^2 df, \quad (52)$$

where $E \{ \cdot \}$ represents the expectation operator and $X_k(f)$ is the Fourier transform of the single batch reference signal $x_k(t)$.

From (50) and (52) and assuming $E \{ x_{cc}^{k(w)}(\tau) (x_{cc}^{l(w)}(\tau))^* \} = 0 \quad \forall l \neq k$, the output power of the noise component is

$$\begin{aligned} E \left\{ \left| M_b^{(w)}(\tau, \nu) \right|^2 \right\} &= \sum_{k=0}^{n_B-1} \sum_{l=0}^{n_B-1} e^{-i2\pi(k-l)\nu T_B} \\ &\quad \times E \left\{ x_{cc}^{k(w)}(\tau) (x_{cc}^{l(w)}(\tau))^* \right\} \\ &= \sum_{k=0}^{n_B-1} E \left\{ \left| x_k^{(w)}(\tau) \right|^2 \right\}. \end{aligned} \quad (53)$$

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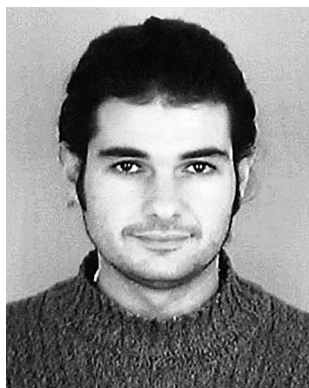
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